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3 Projects	8 Weeks	~360 Hours	Jun 2026

This document summarises the three capstone projects completed during the Bluestock Fintech internship. Each project targets a different domain — geospatial data APIs, financial intelligence dashboards, and mutual fund analytics — while sharing common themes of data engineering, API development, and business intelligence.

For educational purposes only. Not financial advice.

Project 1 — All India Villages API

Status: DONE
Due Date: 30 April 2026
Priority: HIGH

Overview

A production-grade SaaS platform providing a REST API for India's complete village-level geographical data. The platform serves B2B clients needing reliable, standardised address data for drop-down menus and form autocomplete. The hierarchy is: Country → State → District → Sub-District → Village, covering 600,000+ villages.

Tech Stack

Layer	Technology	Purpose
Backend	Node.js + Express.js	REST API, serverless functions
Database	NeonDB (PostgreSQL)	Normalised village data (600K+ rows)
ORM	Prisma	Type-safe queries, migrations
Frontend	React 18 + Vite + TypeScript	Admin panel, B2B portal, demo form
Caching	Redis (Upstash)	Rate limiting, query cache
Auth	JWT + bcrypt	User login + API key auth
Hosting	Vercel	Edge deployment, serverless functions
Charts	Recharts	Admin analytics dashboard

Key Deliverables

- REST API endpoints: /search, /states, /districts, /subdistricts, /villages, /autocomplete
- Admin panel with user management, API key issuance, analytics (bar/line/pie/area charts)
- B2B self-registration portal with API key generation (secret shown once only)
- HMAC authentication, tiered rate limiting (Free: 5K/day → Unlimited: 1M/day)
- Demo contact form showing real-time village autocomplete integration
- Data import pipeline: 600K+ villages from MDDS Excel source into PostgreSQL

API Tiers

Plan	Daily Requests	Price	Access
Free	5,000	$0/month	Single state
Premium	50,000	$49/month	Up to 5 states
Pro	300,000	$199/month	All states
Unlimited	1,000,000	$499/month	All + SLA

Project 2 — Nifty 100 Financial Intelligence (B100)

Status: IN PROGRESS
Due Date: 20 June 2026 (Django) · Power BI: DONE
Priority: HIGH · Progress: 95%

Overview

A full financial intelligence system for India's Nifty 100 companies, covering 12 years of historical data across 100 companies and 7 data tables. The project has three streams: Power BI dashboards (40%), data engineering and Python analytics (35%), and a Django web application with a channel partner API (25%).

Tech Stack

Layer	Technology	Purpose
BI Tool	Power BI Desktop + Service	7 production dashboards
Warehouse	PostgreSQL 15 (star schema)	Dim + fact tables, clean data
ETL	Python + pandas + SQLAlchemy	SQL dump → clean CSV → warehouse
ML	scikit-learn, scipy, statsmodels	Health scoring, anomaly detection, clustering
Web	Django 4.2 + DRF + Chart.js	Public website + 8 chart types per company
Tasks	Celery + Redis	Scheduled ML scoring + ETL refresh
API Auth	HMAC-SHA256 + JWT	Channel partner API security

Power BI Dashboards (7 Total)

Dashboard	Audience	Status
1. Executive Market Overview	Fund managers, CXOs	DONE
2. Company Deep Dive	Retail investors, analysts	DONE
3. Sector Comparison Analyzer	Sector analysts	DONE
4. Financial Health Scorecard	Risk analysts	DONE
5. Growth & Valuation Analytics	Growth investors	DONE
6. Debt & Leverage Monitor	Credit analysts	DONE
7. Dividend & Shareholder Returns	Income investors	DONE

Django Website — Key Pages

- / (Home) — search bar, featured companies, sector cards, AI insights ticker
- /companies/ — filterable & sortable table by ROE, score, sector, health label
- /company/{symbol}/ — 8 Chart.js charts + health score gauge + pros/cons
- /compare/ — side-by-side metric comparison for 2–4 companies
- /screener/ — multi-filter: min ROE, max D/E, sector, health label, sales growth
- /sector/{name}/ — all companies in a sector with combined performance chart

Channel Partner API Rate Limits

Tier	Per Minute	Per Hour	Per Day
BASIC	10	100	500

PRO	60	1,000	10,000
ENTERPRISE	300	10,000	1,00,000

Project 3 — Mutual Fund Analytics

Status: IN PROGRESS
Due Date: 22 June 2026
Priority: MEDIUM · Progress: 70%

Overview

An end-to-end mutual fund analytics platform covering 40 AMFI schemes from January 2022 to December 2025. The project involves data cleaning, a SQLite star-schema warehouse, EDA with 15+ charts, performance metrics (CAGR, Sharpe, Alpha, Beta, VaR), advanced risk analytics, and a 4-page Power BI dashboard.

Tech Stack

Layer	Technology	Purpose
Language	Python 3.11	ETL, analysis, ML
Data	pandas, numpy, scipy	Cleaning, metrics, stats
ML	scikit-learn, statsmodels	Clustering, regression, VaR
Notebooks	Jupyter Notebook	EDA + analytics documentation
Database	SQLite (via SQLAlchemy)	Star schema data warehouse
Viz	matplotlib, seaborn, plotly	15+ EDA charts
BI	Power BI Desktop	4-page interactive dashboard
Report	PDF + PPTX	15–20 page report + 12 slides

Deliverables & Evaluation

ID	Deliverable	Format	Weight
D1	ETL pipeline	.py script	15%
D2	SQLite database	.db file	10%
D3	EDA notebook	.ipynb	15%
D4	Performance metrics	.ipynb + CSVs	15%
D5	Power BI dashboard	.pbix	20%
D6	Advanced analytics	.ipynb	10%
D7	Final report + slides	.pdf + .pptx	15%

Key Analytics Tasks

- Data cleaning: 10 CSVs — NAV history, transactions, scheme performance, AUM, holdings
- EDA: NAV trends 2022–2026, AUM by fund house, SIP inflows, folio count growth (13 Cr → 26 Cr)
- Performance: CAGR (1yr/3yr/5yr), Sharpe Ratio, Sortino Ratio, Alpha & Beta (OLS regression)
- Risk: Historical VaR (95%), CVaR, Maximum Drawdown, rolling 90-day Sharpe
- Advanced: Investor cohort analysis, SIP continuity flags, fund recommender by risk grade
- Dashboard: 4 pages — industry overview, fund performance, investor analytics, SIP trends
- Final report: 15–20 page PDF + 12-slide presentation

Tools & Setup Guide (All Free)

Tool	Download From	Used In
Python 3.11	python.org	Projects 2, 3
VS Code	code.visualstudio.com	All projects
Power BI Desktop	microsoft.com/en-us/power-bi	Projects 2, 3
PostgreSQL 15	postgresql.org	Projects 1, 2
DBeaver (DB viewer)	dbeaver.io	Projects 1, 2, 3
Node.js (LTS)	nodejs.org	Project 1
Git	git-scm.com	All projects
Postman	postman.com	Projects 1, 2
Jupyter Notebook	pip install jupyter	Projects 2, 3
NeonDB (free tier)	neon.tech	Project 1
Vercel (free)	vercel.com	Project 1

Quick Start Commands

Install Python packages (Projects 2 & 3):

```
pip install pandas sqlalchemy psycopg2-binary django djangorestframework scikit-learn jupyter
matplotlib seaborn plotly
```

Install Node packages (Project 1):

```
npm install express prisma @prisma/client redis jsonwebtoken bcrypt
```

Start Django dev server (Project 2):

```
python manage.py runserver
```

Open Jupyter notebooks (Project 3):

```
jupyter notebook
```

Start Celery worker (Project 2):

```
celery -A bluestock worker -l info
```